

Parth Sanjay Nimbalkar

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EDUCATION

University at Buffalo, SUNY
Master of Science, Computer Science

Buffalo, NY
Aug. 2023 – Feb 2025

University of Mumbai
Bachelor of Engineering, Computer Engineering

Mumbai, India
Aug. 2019 – May 2023

EXPERIENCE

Software Engineer
Crowddoing

Feb 2025 – Present
San Francisco, CA

- Developed a **Wellbeing Module** in **Java** for **Android** with live activity tracking, delivering insights to **5,000+ users**.
- Implemented **real-time data sync** via **Firebase** with **offline caching**, conflict resolution, ensuring **99% data integrity**.
- Integrated **AWS DynamoDB** as backend storage for high-availability writes, managing **500K+ activity records**.
- Collaborated with stakeholders on wellbeing features in bi-weekly **Agile** releases, **boosting adoption by 15%**.

Software Engineer Intern
Little Angel Foundation

Jul 2022 – Jul 2023
Navi Mumbai, India

- Built a **Java microservice** with **PostgreSQL** to **track 10K+ donor records** and **cut manual data errors by 60%**.
- Designed **API endpoints** and query logic for donor activity, **saving 5+ hours per week** for internal reporting teams.
- Engineered **Spring Batch ETL workflows** to **ingest 100K+ legacy records**, reducing **processing time by 70%**.
- Deployed the donor management service on **AWS EC2** with environment configs, **improving system uptime to 99%**.

Frontend Developer Intern
TCR Innovation

Apr 2022 – Jun 2022
Navi Mumbai, India

- Redesigned **React** components for client dashboards, improving UI load time by **40%** and reducing redundancy by **30%**.
- Integrated secure **Spring Boot APIs** to render dynamic account data in React, **reducing response latency by 35%**.
- Implemented **OAuth 2.0 authentication** with secure session handling and token validation, enabling role-based access.
- Refactored shared UI logic into modular patterns, boosting frontend scalability and accelerating **feature rollout by 25%**.

PROJECTS

Taco-DB – Relational Database Engine | *C++, POSIX, Linux, Bash, GoogleTest*

- Engineered a disk-based **heap file and buffer manager** to cache pages and **boost I/O throughput** for large datasets.
- Implemented **B+ Tree indexing** and **join algorithms** to **speed up query access paths** and join operations.
- Built **relational operators** and **tuned query plans** for **TPC-H workloads**, improving **runtime efficiency by 30%**.

PintOS – Operating System Kernel | *C, GDB, OOPS*

- Designed an **alarm clock subsystem** using **timer interrupts** to block/wake threads efficiently **without busy-waiting**.
- Built **syscall** and **page fault handlers** for **process control** and **memory protection** to **ensure user-kernel isolation**.
- Devised a **multi-level feedback queue scheduler (MLFQS)** to control **thread priorities** and **optimize CPU usage**.

DistriCache – Distributed Key - Value Store | *Java, gRPC, Multithreading, Consistent Hashing, Write-Ahead Logging*

- Designed a **distributed key-value store** with consistent hashing for **fault-tolerant data partitioning across 3 nodes**.
- Implemented **passive replication** and **write-ahead logging** for **consistency** and **recovery from node failures**.
- Built **multithreaded request handlers** using **gRPC** to simulate inter-node communication and concurrent client access.

Unicare – Secure Health Record Platform | *Java, Spring Boot, Solidity, React, IPFS, Amazon RDS, JWT*

- Built a **decentralised EHR system** using **Spring Boot** and **React** to ensure **continuous availability** of patient data.
- Wrote **smart contracts** in **Solidity** to enforce **role-based access** and **auditability** on the **Ethereum blockchain**.
- Orchestrated **secure hybrid data access** using **IPFS**, **Amazon RDS**, and **JWT-protected Spring Boot APIs**.

TECHNICAL SKILLS

Languages: Java, C++, C, Python, SQL

Databases: PostgreSQL, MySQL, MongoDB

Data Engineering: Spring Batch, ETL Pipelines, Data Warehousing, Query Optimization

Cloud and DevOps: AWS (EC2, RDS, Redshift, Lambda, DynamoDB), Docker, Kubernetes, Git, CI/CD

Distributed Systems: Multithreading, Concurrency, Microservices, gRPC

Web and Security: Spring Boot, React, RESTful APIs, JWT, RBAC